

NGUYEN KHAC BAO

AI Engineer Intern

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CAREER OBJECTIVE

AI Engineer Intern candidate with hands-on experience building RAG/GenAI systems, fine-tuning LLMs, and developing backend data pipelines. Seeking to apply a Software Engineering mindset to build trustworthy AI applications for complex domains.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

2022 – 2027

Bachelor of Information Technology

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash, C/C++ (basic)

GenAI / LLMs: RAG pipelines, Agentic RAG, LlamaIndex ReAct tools, PEFT/LoRA, Llama 3.1, Prompt Engineering, Gemini/OpenAI-compatible APIs, llama.cpp, GGUF

Backend / Infrastructure: FastAPI, Docker, Docker Compose, Celery, Valkey/Redis, MariaDB, Git/GitHub, Linux, Vast.ai

Data / Vector DB: Qdrant, Hybrid Retrieval, Reranking, Pandas, Hugging Face, JSONL, Dataset QA

Computer Vision: PyTorch, Ultralytics YOLOv8/v11, YOLO-Pose, OpenCV, ByteTrack, HyperLPR3/OCR

LANGUAGES

TOEIC: 610/990

2026

English: Technical reading proficiency

Able to read official documentation, research papers, model cards, API references, and engineering articles in English.

PROJECTS

Vietnamese Legal RAG Chatbot

2026

Personal Project | Technology: Python, FastAPI, Streamlit, Celery, Valkey, MariaDB, Qdrant, PEFT/LoRA, Llama 3.1, llama.cpp | Source: [GitHub](#)

- Built an end-to-end Vietnamese legal assistant with a FastAPI backend, Streamlit chat UI, Celery asynchronous task handling, Valkey/Redis queueing, and MariaDB conversation storage.
- Designed an agent-style RAG workflow to route user queries between legal retrieval, LlamaIndex ReAct tools, web-search handling, and chat fallback.
- Implemented hybrid retrieval over Qdrant vector database, multi-query expansion, reranking, and legal-context prompting for Vietnamese legal QA.
- Fine-tuned Llama-3.1-8B-Instruct with PEFT/LoRA on Vietnamese legal QA data using Vast.ai; served the adapter through an OpenAI-compatible local API.
- Optimized local inference by converting the serving path to llama.cpp with GGUF Q4, reducing memory pressure compared with full FP16 Transformers inference.

FallGuard Labeling Tool – Computer Vision Dataset Assistant

2026

Personal Project | Technology: Python, Streamlit, OpenCV, Pandas, YOLO-Pose, PyYAML, pytest | Source: [GitHub](#)

- Built a Python-based dataset assistant for fall-detection image/video datasets, covering raw data import, video frame extraction, metadata generation, human review, QA checks, split, and export.
- Integrated YOLO-Pose to generate 13-point human pose annotations and built a Streamlit UI for label correction, uncertain/excluded samples, pose review, and manual keypoint editing.
- Designed reproducible CSV source-of-truth records using metadata, annotations, keypoints, and annotation events; tested with Kaggle Fall Detection and Multiple Cameras Fall datasets.

Traffic Violation Detection & License Plate Recognition

2026

Personal Project | Technology: Python, OpenCV, Ultralytics YOLO, ByteTrack, HyperLPR3, NumPy, Shapely, JSON | Source: [GitHub](#)

- Built a fixed-camera traffic video pipeline for vehicle detection, multi-object tracking, traffic-light state estimation, stop-line violation logic, license plate detection, OCR, and evidence generation.
- Fine-tuned a YOLO-based license plate detector on a compact CCPD2019 subset of 8,598 images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50 on validation.
- Implemented dataset QA for broken files, YOLO bbox validity, negative sample logic, OCR crop availability, duplicate review, low-light images, heavy blur, and small license plates.

ACTIVITIES

AI Research Group – University Club

2025 – Present

Member

Discussed Machine Learning, Deep Learning, Computer Vision, and AI application development.